



Semester Gasal 2025-2026

Pembelajaran Mesin "Machine Learning" Courses

20 Agustus 2025

START

Dr. Cucun Very Angkoso, MT



Welcome

Kelas Pembelajaran Mesin

Semester Gasal 2025-2026





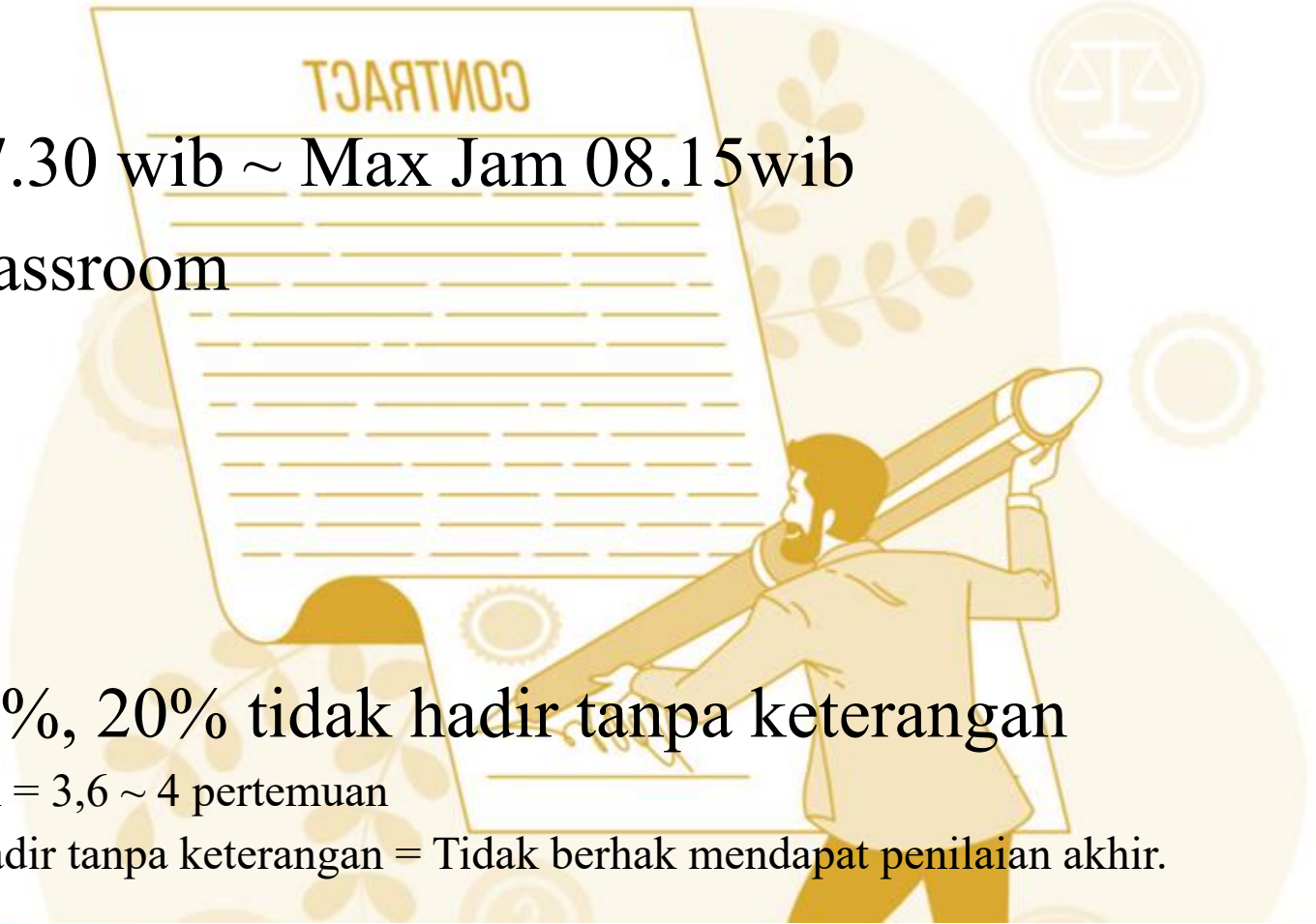
Course Information

- Course Title : Machine Learning / “*Pembelajaran Mesin*”
- Course Code : ML-IF 101
- Semester : 2025-2026 (1)
- Credits : 3
- Instructor : Dr. Cucun Very Angkoso, MT
- Email : cucunvery@ieee.com
- G-Classroom : (B) zmgrvfgc / (C) mzybgy6d

next slide →

Kontrak Kuliah

- Perkuliahan dimulai 07.30 wib ~ Max Jam 08.15wib
- Wajib bergabung G-Classroom
- Penilaian akhir
 - UAS : 30%
 - UTS : 30%
 - Tugas : 40%
- Absensi : threshold 80%, 20% tidak hadir tanpa keterangan
 - $20\% \times 16 \text{ Pertemuan} = 3,6 \sim 4 \text{ pertemuan}$
 - 4 pertemuan tidak hadir tanpa keterangan = Tidak berhak mendapat penilaian akhir.



Course Description



- Memperkenalkan konsep dan teknik dasar di bidang pembelajaran mesin kepada mahasiswa.
- Meliputi landasan teori dan implementasi praktis, kursus ini akan membekali mahasiswa dengan keterampilan yang dibutuhkan untuk merancang, mengimplementasikan, dan mengevaluasi model pembelajaran mesin.
- Mahasiswa akan mengeksplorasi berbagai algoritma ML, mempelajari cara melakukan *Preprocessing Data*, dan menerapkan metode pembelajaran mesin pada masalah dunia nyata.
- Membahas topik-topik lanjutan seperti jaringan saraf, pemrosesan bahasa alami, dan visi komputer, beserta diskusi tentang pertimbangan etika dalam AI.



Learning Outcomes

- **Memahami Konsep Dasar:** Menetapkan dan membedakan antara berbagai jenis pembelajaran mesin : supervised, unsupervised, reinforcement learning dan aplikasinya.
- **Memroses dan Menganalisis Data:** Menerapkan teknik untuk pembersihan data, normalisasi, dan analisis data eksploratori untuk menyiapkan kumpulan data untuk pelatihan model.
- **Mengimplementasikan Algoritma Pembelajaran Mesin:** Mengembangkan dan mengimplementasikan model regresi dan klasifikasi dasar menggunakan algoritma seperti regresi linier, regresi logistik, pohon keputusan, dan support vector machine.
- **Mengevaluasi Kinerja Model:** Menggunakan validasi silang dan metrik evaluasi lainnya untuk menilai kinerja model pembelajaran mesin, memahami trade-off bias-varians.
- **Menerapkan Pembelajaran Tak Terbimbing:** Memanfaatkan teknik pengelompokan dan pengurangan dimensionalitas seperti k-means dan PCA untuk mengungkap pola dalam data tanpa hasil yang diberi label.



Learning Outcomes

- **Menjelajahi Topik Lanjutan:** Memperoleh pengetahuan pengantar tentang metode ensemble, jaringan neural, dan pembelajaran mendalam, serta memahami aplikasinya dalam pemecahan masalah yang kompleks.
- **Bekerja dengan Pustaka Pembelajaran Mesin:** Memanfaatkan pustaka Python seperti Scikit-learn, TensorFlow, dan Keras untuk membangun dan menerapkan model pembelajaran mesin.
- **Menyelidiki Aplikasi Khusus:** Menerapkan teknik pembelajaran mesin pada bidang khusus seperti pemrosesan bahasa alami dan visi komputer.
- **Mendiskusikan Implikasi Etis:** Mengartikulasikan pertimbangan etika dalam pembelajaran mesin, termasuk bias, privasi, dan dampak sosial dari teknologi AI.
- **Melakukan Proyek Pembelajaran Mesin:** Berhasil menyelesaikan proyek yang melibatkan pengumpulan data, pelatihan model, validasi, dan penerapan, yang menunjukkan kemampuan untuk menerapkan konsep yang telah dipelajari ke dalam skenario praktis.



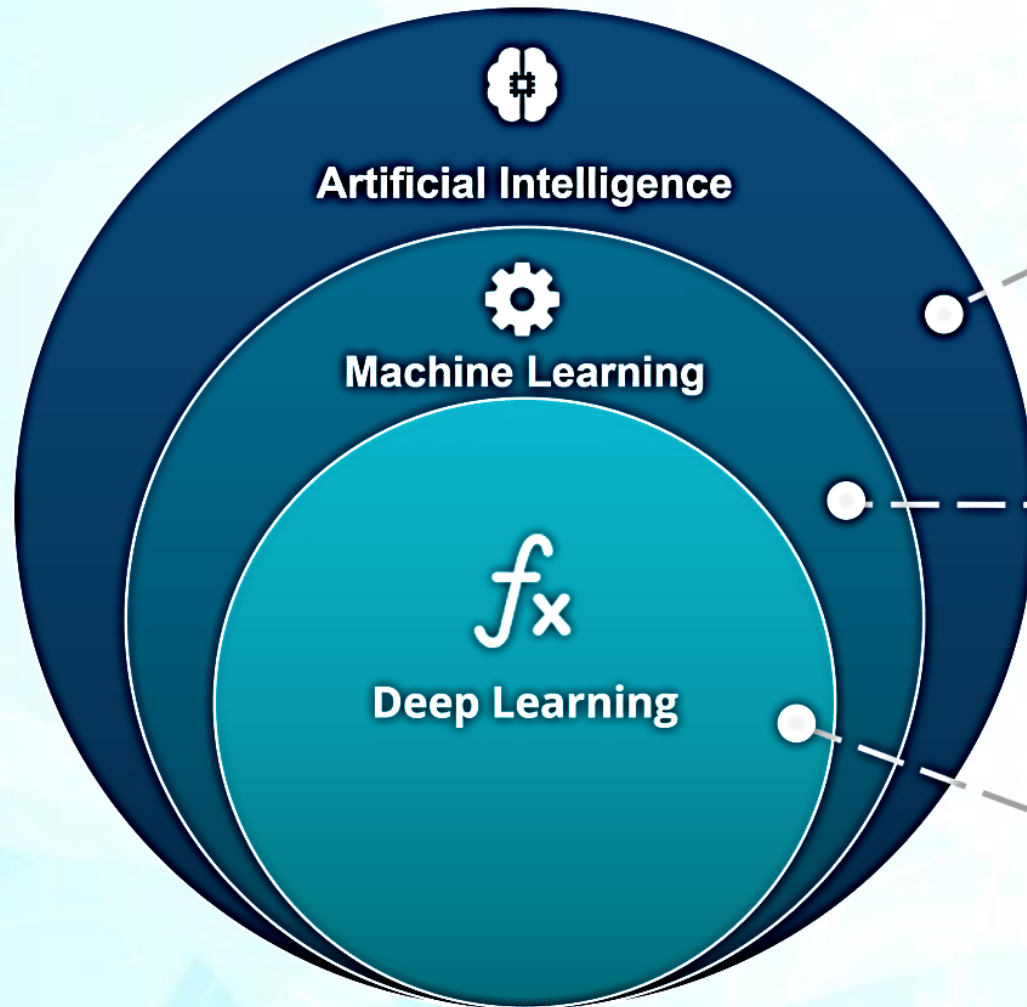
Materi Perkuliahan

- **Topic 1-2: Introduction to Machine Learning**
 - **Overview of Machine Learning:** Definitions, history, and types (supervised, unsupervised, reinforcement learning) [1](#)
 - **Applications:** Real-world applications in various industries.
- **Topic 3-4: Data Preprocessing and Exploration**
 - **Data Cleaning and Preparation:** Handling missing data, normalization, and feature scaling.
 - **Exploratory Data Analysis (EDA):** Visualizing data, understanding distributions, and identifying patterns.
- **Topic 5-6: Supervised Learning**
 - **Regression Techniques:** Linear regression, polynomial regression, and evaluation metrics like MSE, R-squared.
 - **Classification Algorithms:** Logistic regression, decision trees, and support vector machines.
- **Topic 7-8: Unsupervised Learning**
 - **Clustering Methods:** K-means, hierarchical clustering, and DBSCAN.
 - **Dimensionality Reduction:** Principal Component Analysis (PCA) and Singular Value Decomposition (SVD).
- **Topic 9: Model Evaluation and Selection**
 - **Cross-Validation:** Techniques like k-fold cross-validation.
 - **Bias-Variance Trade-off:** Understanding overfitting and underfitting.



Materi Perkuliahan

- **Topic 10: Advanced Topics**
 - **Ensemble Methods:** Bagging, boosting, and random forests.
 - **Neural Networks and Deep Learning:** Basics of neural networks, introduction to deep learning.
- **Topic 11: Practical Implementation**
 - **Programming with Python:** Using libraries like Scikit-learn, TensorFlow, and Keras for building models.
 - **Project Work:** Implementing a machine learning project from data collection to model deployment.
- **Topic 12: Special Topics and Applications**
 - **Natural Language Processing (NLP):** Basics of text processing and sentiment analysis.
 - **Computer Vision:** Introduction to image processing and convolutional neural networks (CNNs).
- **Topic 13: Ethical Considerations and Future Trends**
 - **Ethics in AI and ML:** Bias in algorithms, privacy concerns, and ethical AI practices.
 - **Future of Machine Learning:** Emerging trends and technologies in AI and ML.
- **Topic 14: Review and Final Exam Preparation**
 - **Review Sessions:** Recap of key concepts and techniques.
 - **Exam Preparation:** Practice problems and mock exams.



ARTIFICIAL INTELLIGENCE

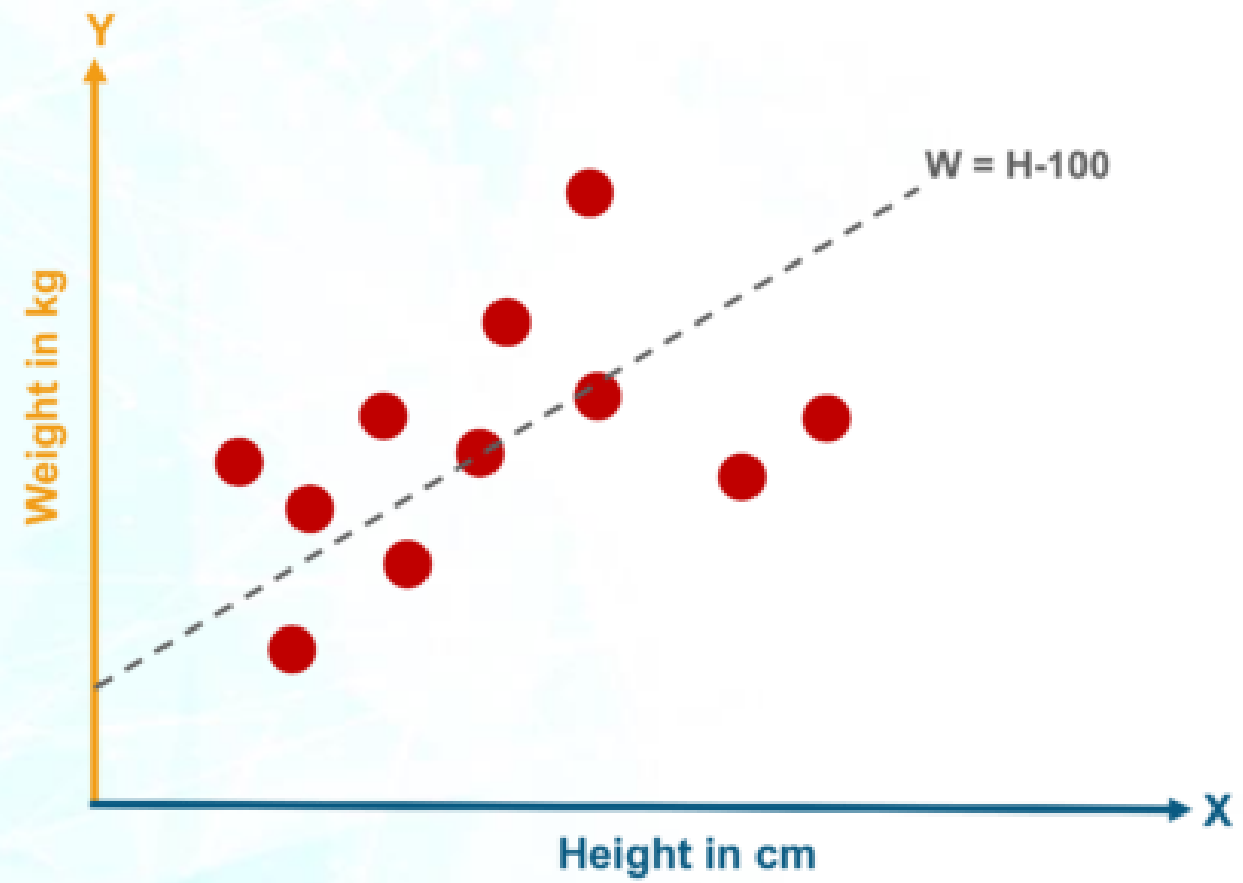
A technique which enables machines to mimic human behaviour

MACHINE LEARNING

Subset of AI technique which use statistical methods to enable machines to improve with experience

DEEP LEARNING

Subset of ML which make the computation of multi-layer neural network feasible





Apa Itu Pembelajaran Mesin?

- Automating automation
- Getting computers to program themselves
- Writing software is the bottleneck
- Let the data do the work instead!
- Consider using machine learning when:
 - Have **a complex task or problem** involving:
 - A large amount of data
 - Lots of variables
 - But, **no existing formula or equation.**



Apa Itu Pembelajaran Mesin?

- “Learning is any process by which a system improves performance from experience.”
- “Pengalaman sebelumnya dapat mempengaruhi cara individu memahami dan memproses informasi baru.”
- Herbert Simon
- Definisi oleh Tom Mitchell (1998):
- Pembelajaran Mesin adalah studi tentang algoritma untuk meningkatkan kinerjanya **P** pada beberapa tugas **T** melalui beberapa pengalaman **E**.
- Tugas pembelajaran didefinisikan sebagai **<P,T,E>**.

Machine Learning Model

- Machine learning ditujukan untuk mendapatkan informasi pola data di dalam dataset untuk selanjutnya digunakan untuk menyelesaikan suatu tugas tertentu.
- Algoritma yang dapat belajar dari suatu dataset pelatihan

Traditional programming



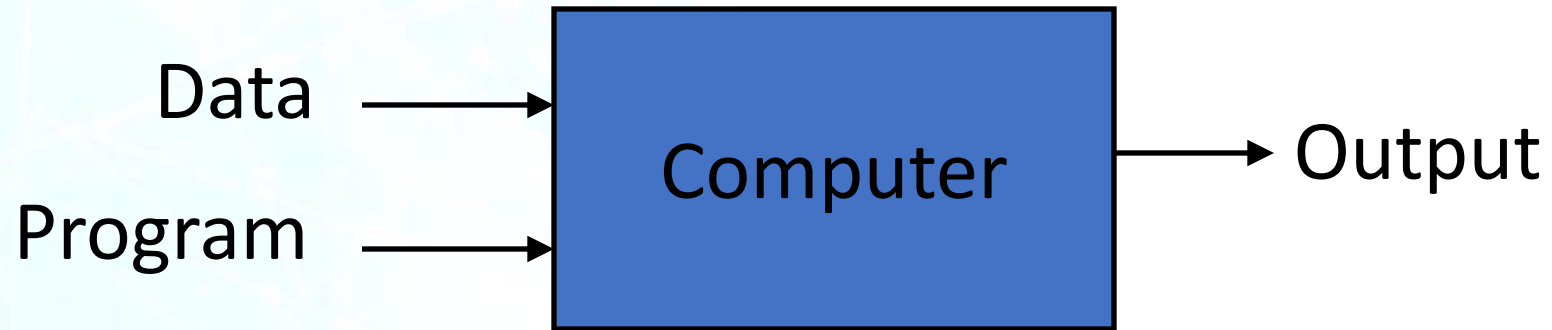
Machine learning



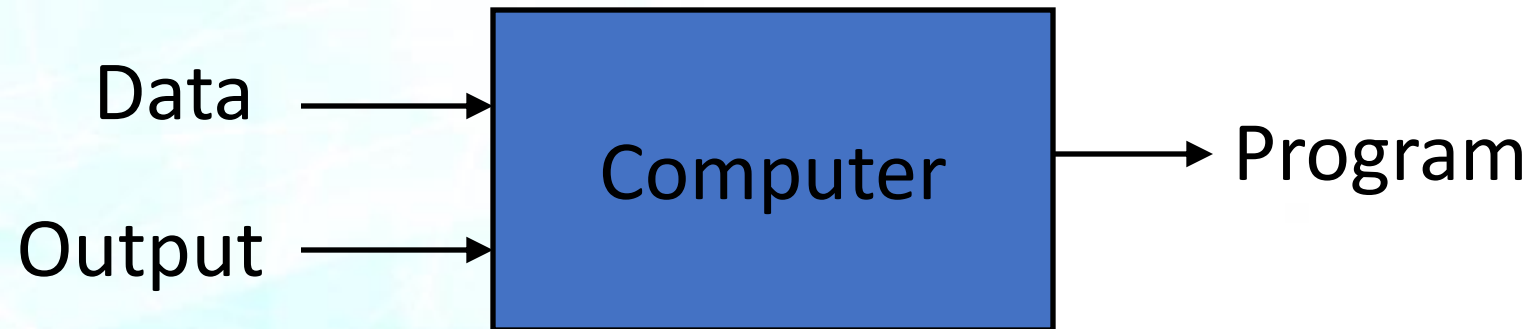
**WHILE TRUE:
LEARN()**

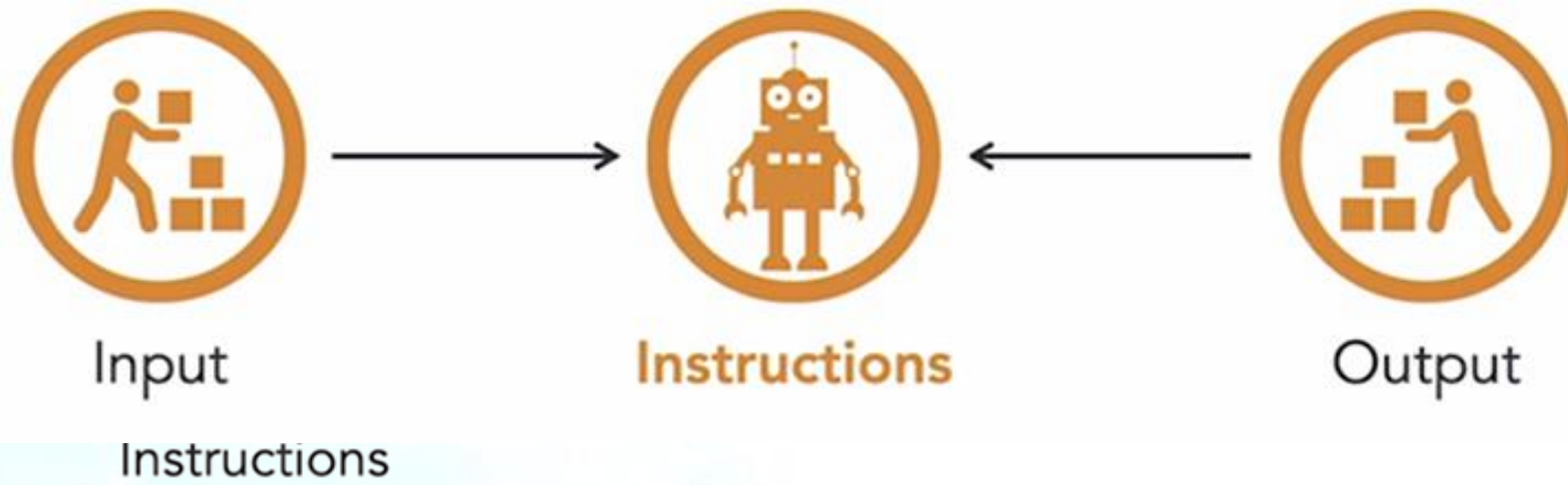


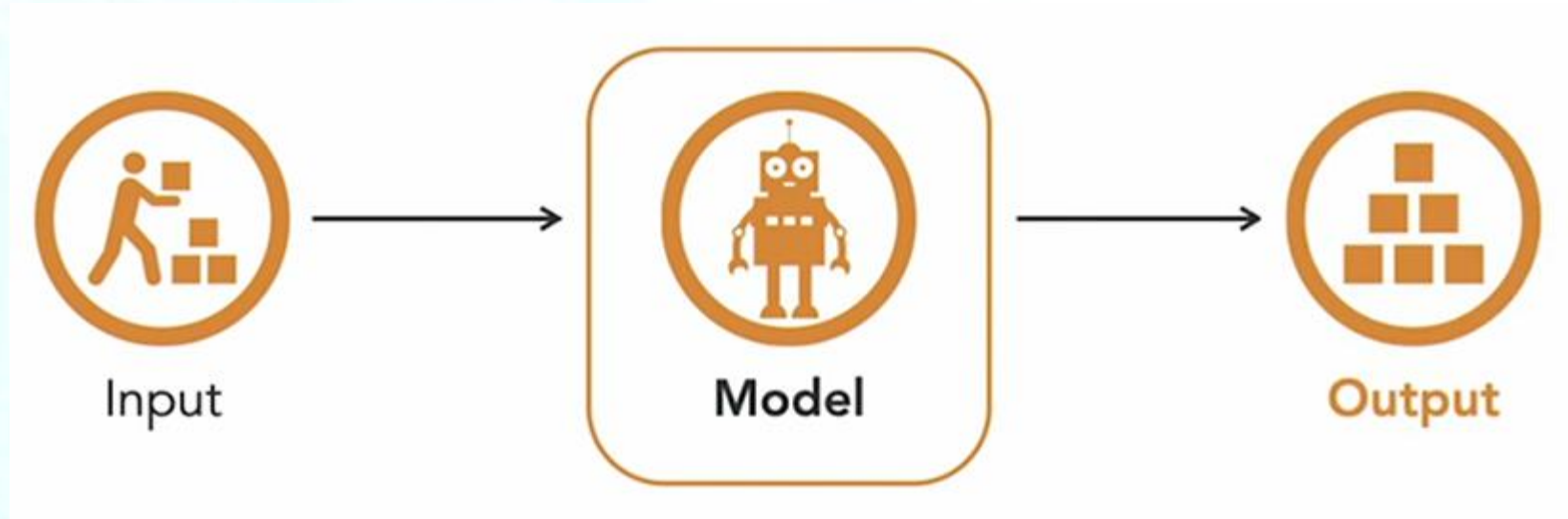
Pemrograman tradisional



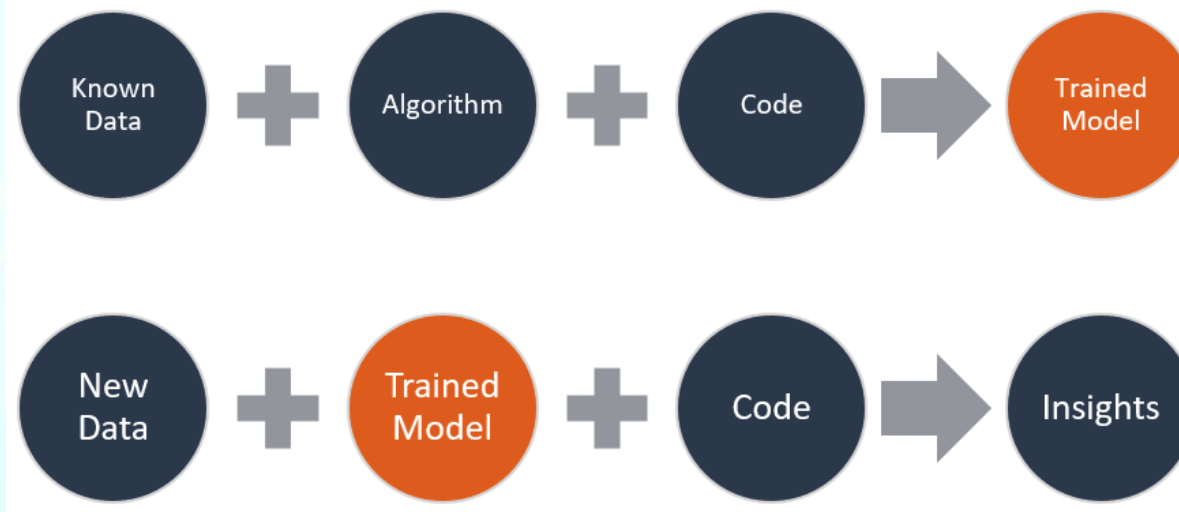
Pembelajaran mesin (*Machine Learning*)





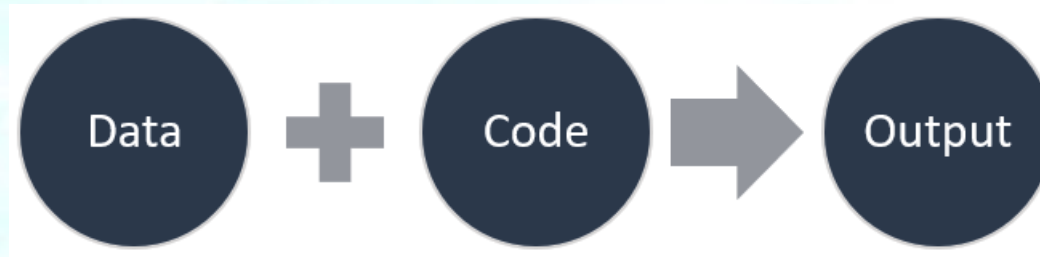


Input proses & output

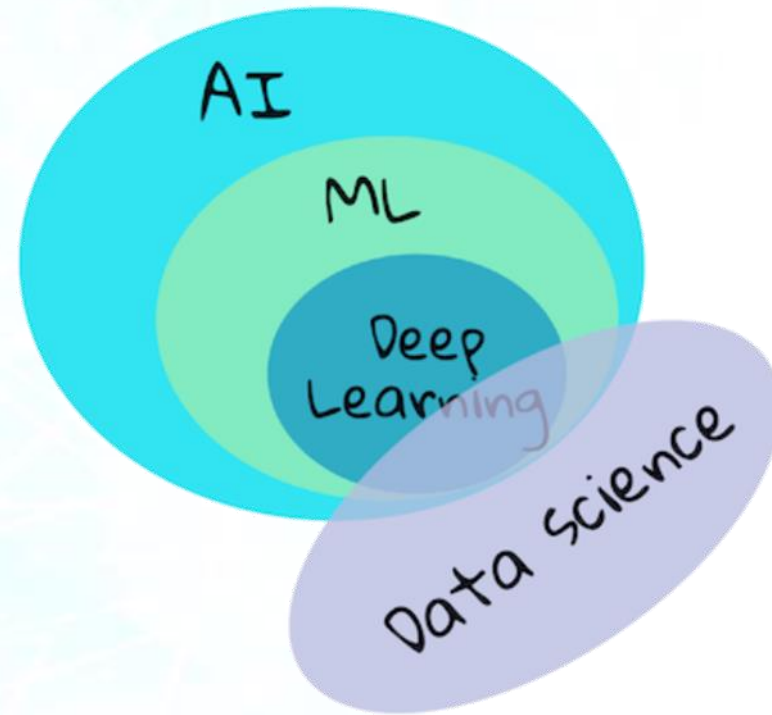


pekerjaan Data science

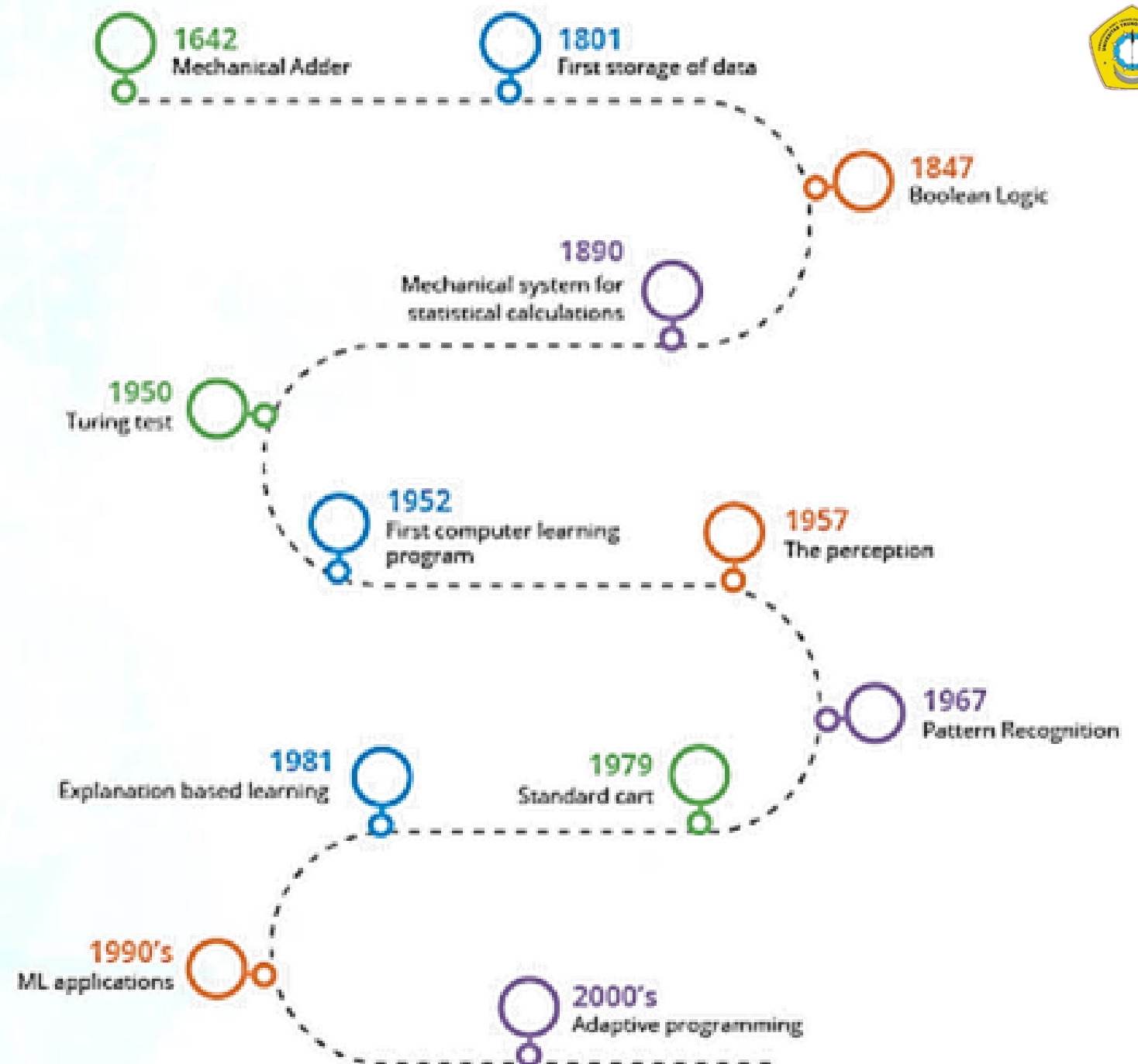
Berbeda dengan rekayasa perangkat lunak coding dilakukan terhadap spesifikasi dengan menghasilkan keluaran yang sama untuk data masukan yang sama setiap kali kode dijalankan



pekerjaan
rekayasa perangkat lunak



History of Machine Learning and AI



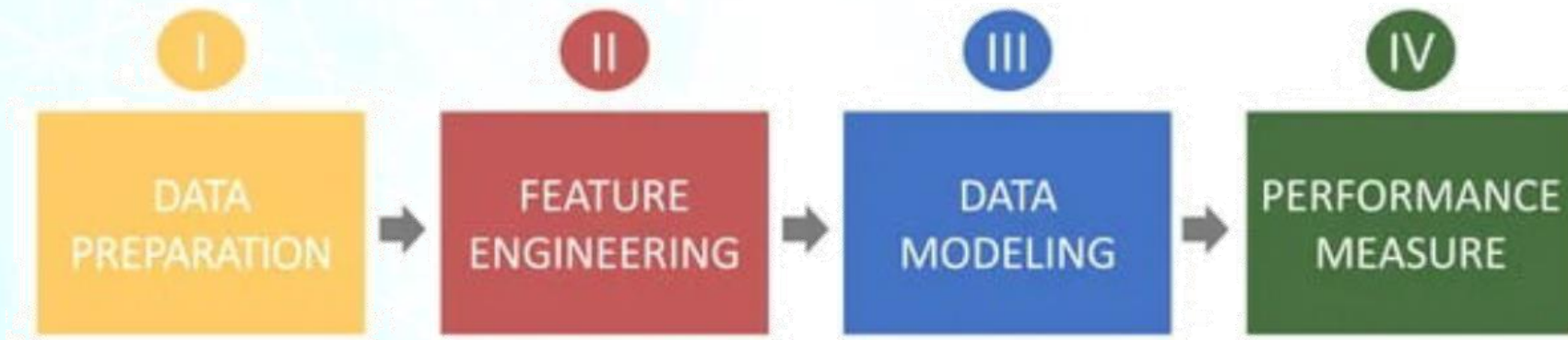


Ekplorasi tool ML





Langkah pengembangan ML-Model



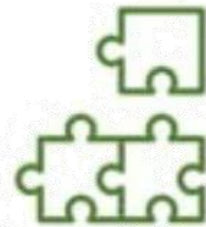
DATA

+



ALGORITHMS

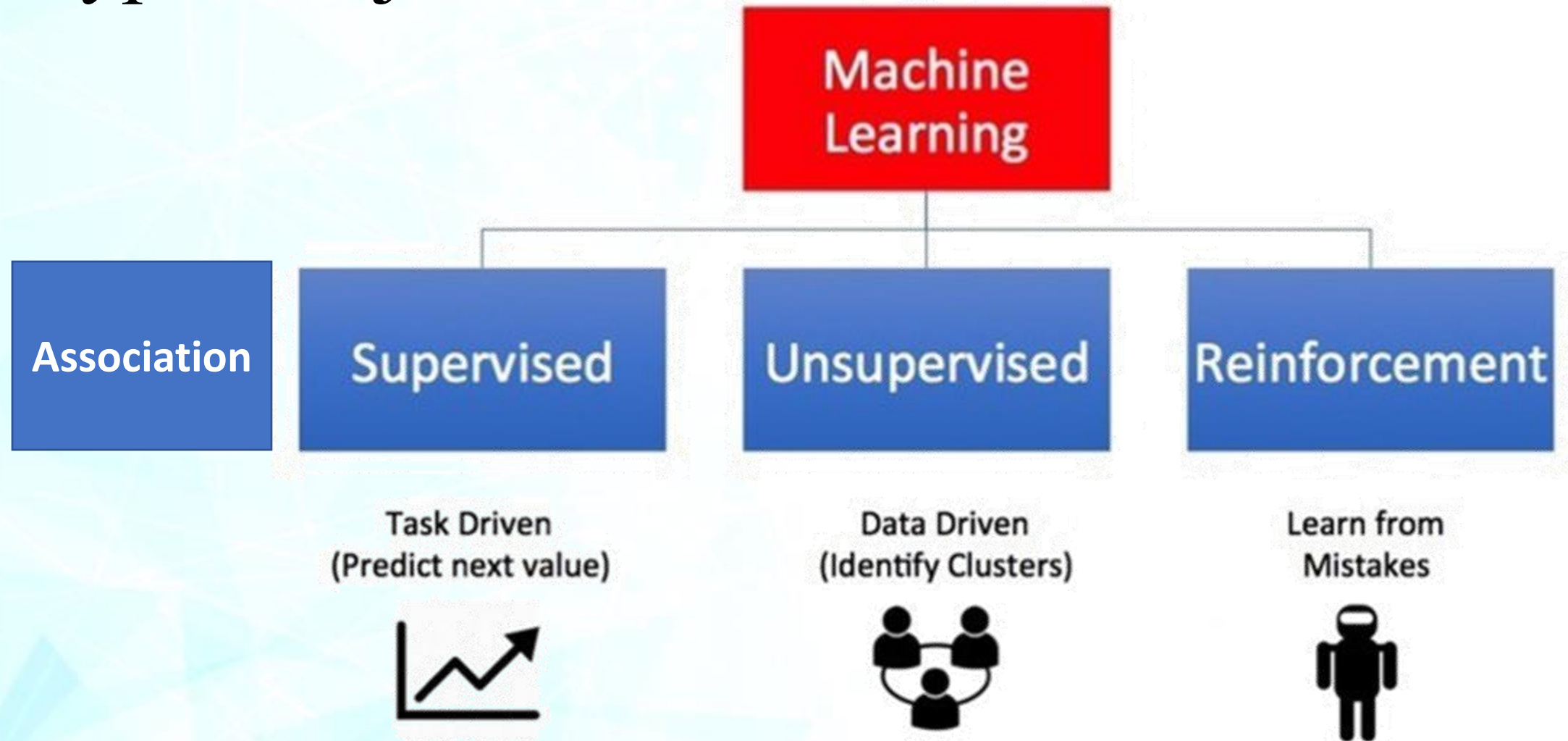
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MODEL



Type Belajar Mesin





Learning Associations

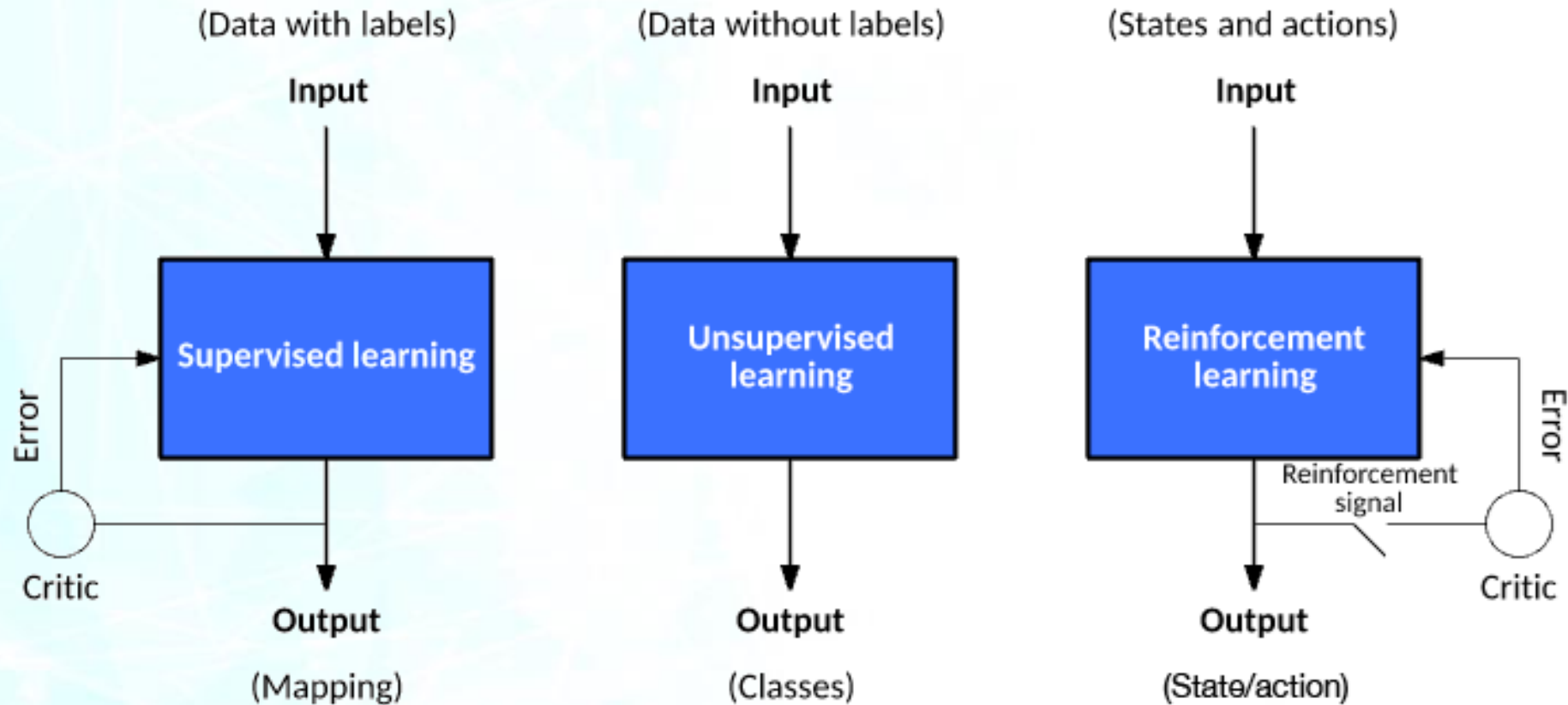
- Basket analysis:

$P(Y | X)$ probability that somebody who buys X also buys Y where X and Y are products/services.

Example: $P(\text{chips} | \text{beer}) = 0.7$



Type Belajar Mesin





Type Belajar Mesin

supervised learning

Input data



Annotations

These are
apples



Model

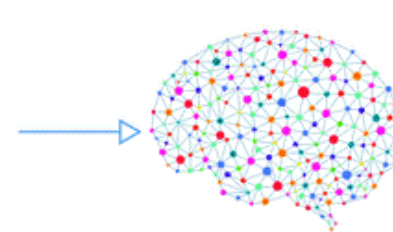


Prediction

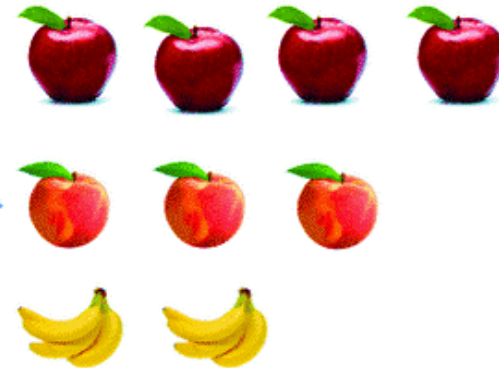
Its an
apple!

unsupervised learning

Input data



Model





Supervised Learning

Independent Variables

Training
Data

Customer	Amount	Grade	Purpose
James S.	\$10,000	C	Business
Carol J.	\$40,000	B	Home
Sarah P.	\$15,000	A	Car
John J.	\$20,000	B	Home
Saul B.	\$30,000	A	Business
Alan Q.	\$40,000	A	Medical
Frank D.	\$5,000	B	Vacation
Jane R.	\$12,000	A	Car
Bill T.	\$35,000	C	Home
Tom C.	\$15,000	B	Medical



Default

Yes

Yes

No

No

No

No

Yes

No

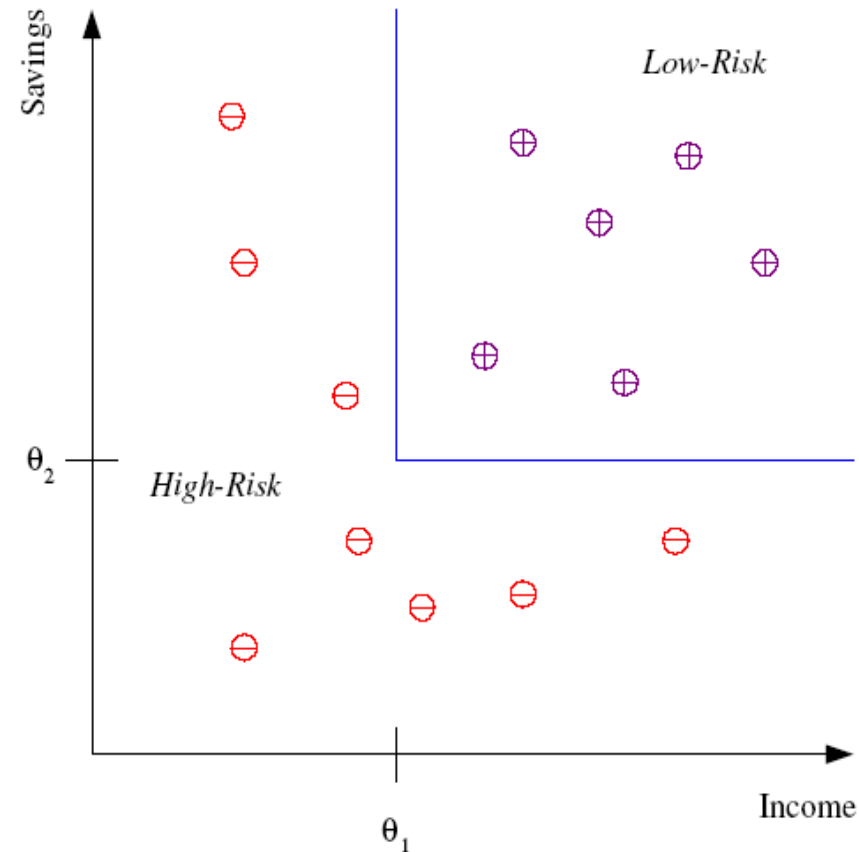
No

Yes

Dependent Variable

Classification

- Example: Credit scoring
- Differentiating between **low-risk** and **high-risk** customers from their *income* and *savings*



Discriminant: IF $income > \theta_1$ AND $savings > \theta_2$
THEN **low-risk** ELSE **high-risk**



Classification: Applications

- Aka Pattern recognition
- **Face recognition:** Pose, lighting, occlusion (glasses, beard), make-up, hair style
- **Character recognition:** Different handwriting styles.
- **Speech recognition:** Temporal dependency.
 - Use of a dictionary or the syntax of the language.
 - Sensor fusion: Combine multiple modalities; eg, visual (lip image) and acoustic for speech
- **Medical diagnosis:** From symptoms to illnesses
- ...



Face Recognition

Training examples of a person



Test images



Regression

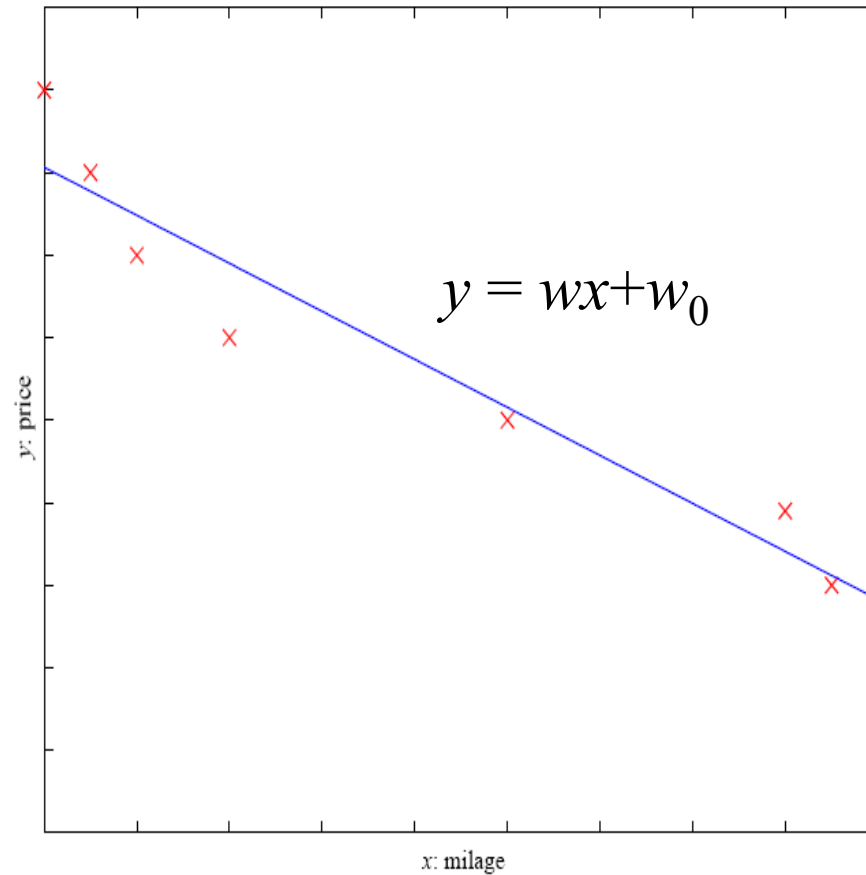
- Example: Price of a used car
- x : car attributes

y : price

$$y = g(x | \theta)$$

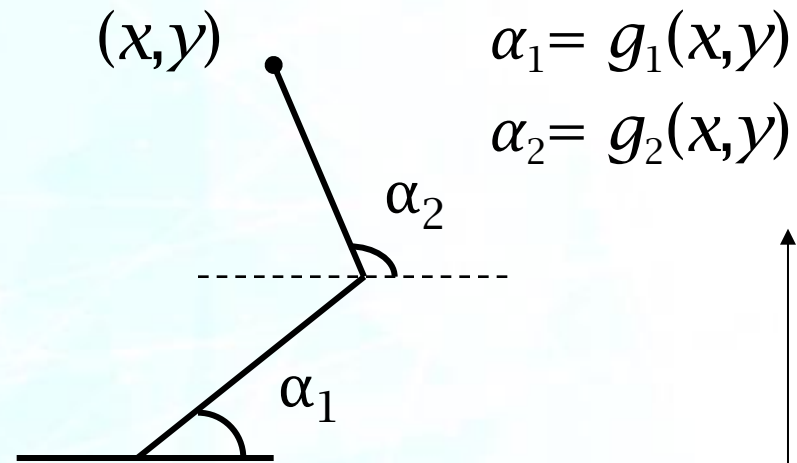
$g()$ model,

θ parameters

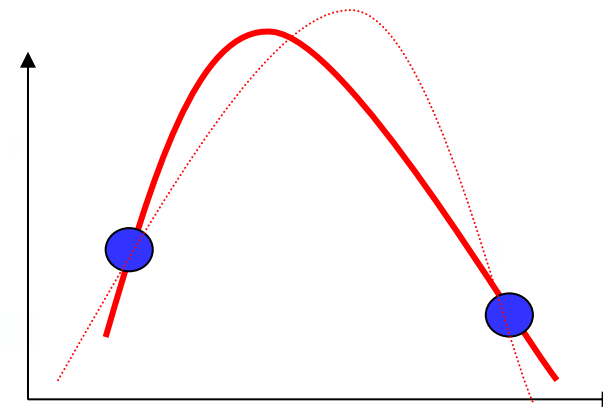


Regression Applications

- Navigating a car: Angle of the steering wheel (CMU NavLab)
- Kinematics of a robot arm



■ Response surface design





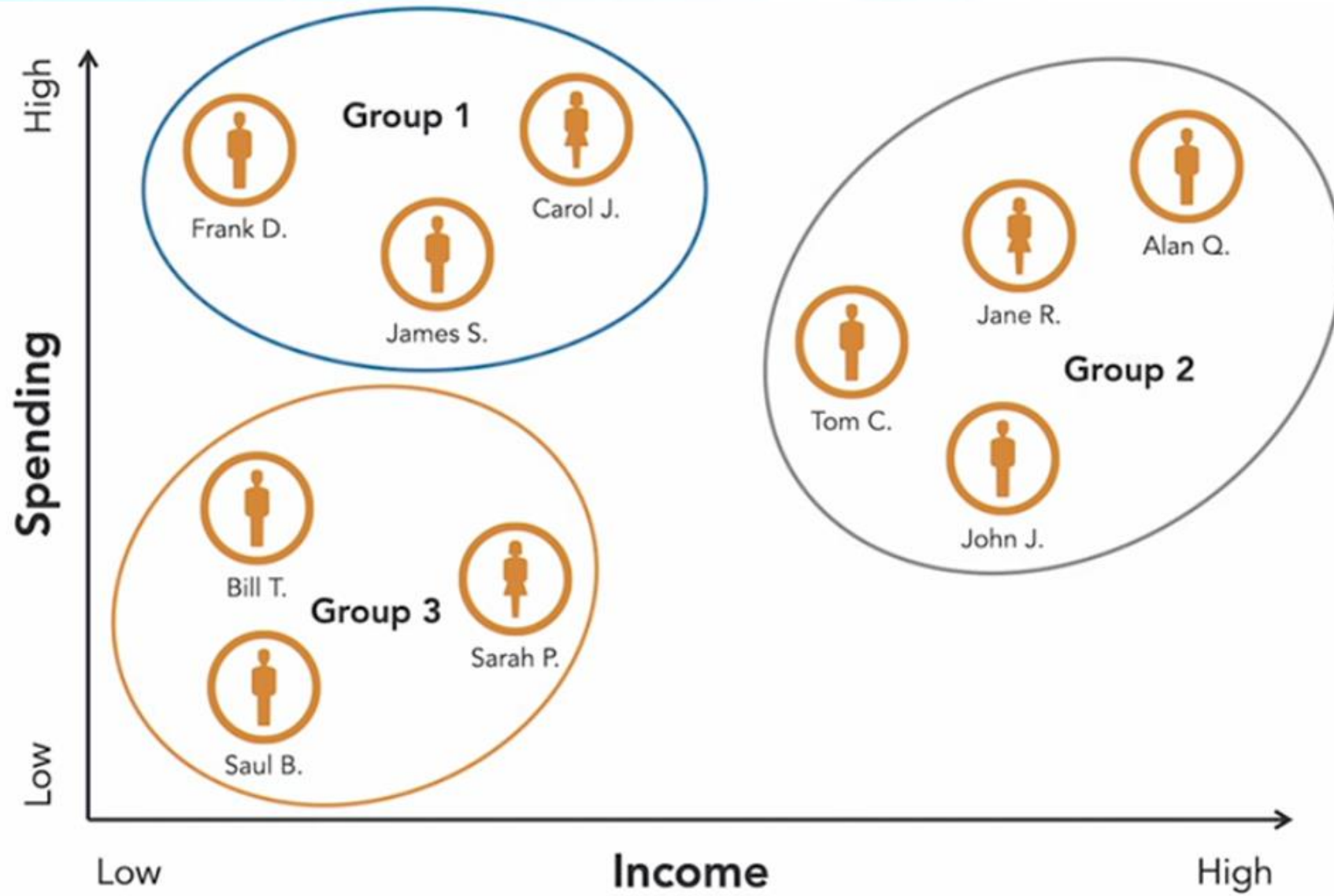
Supervised Learning: Uses

- **Prediction of future cases:** Use the rule to predict the output for future inputs
- **Knowledge extraction:** The rule is easy to understand
- **Compression:** The rule is simpler than the data it explains
- **Outlier detection:** Exceptions that are not covered by the rule, e.g., fraud



Unsupervised Learning

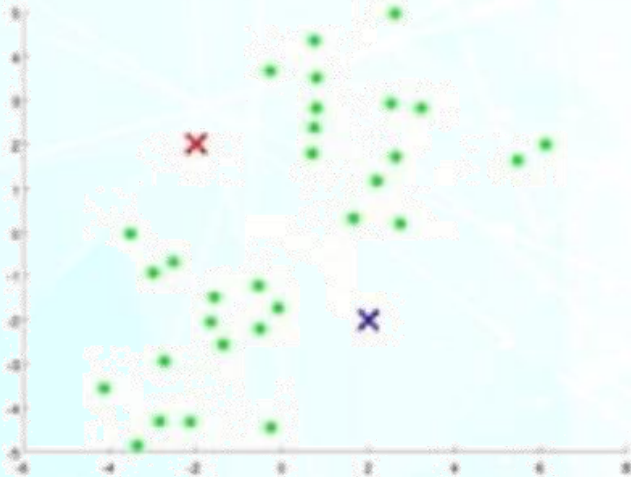
- Learning “"apa yang biasanya terjadi"”
- No output
- Clustering: Mengelompokkan insobyek/data yang serupa
- Contoh aplikasi :
 - Customer segmentation in CRM
 - Image compression: Color quantization
 - Bioinformatika: Motif pembelajaran





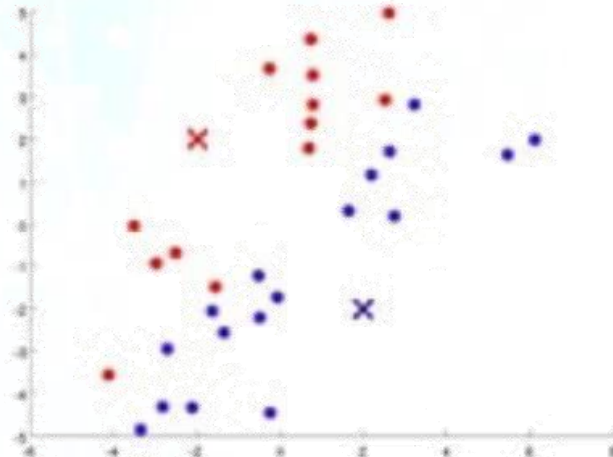
Customer	Spending	Income	Age	Gender	Education	Segment
James S.	7	\$40,000	22	M	High School	Group 1
Carol J.	10	\$50,000	40	F	Bachelors	Group 1
Sarah P.	2	\$36,000	53	F	Bachelors	Group 3
John J.	3	\$72,000	24	M	Doctorate	Group 2
Saul B.	1	\$18,000	18	M	High School	Group 3
Alan Q.	9	\$98,000	32	M	Doctorate	Group 2
Frank D.	8	\$28,000	19	M	High School	Group 1
Jane R.	6	\$80,000	47	F	Bachelors	Group 2
Bill T.	4	\$16,000	25	M	High School	Group 3
Tom C.	5	\$63,000	62	M	Masters	Group 2

Unsupervised Learning



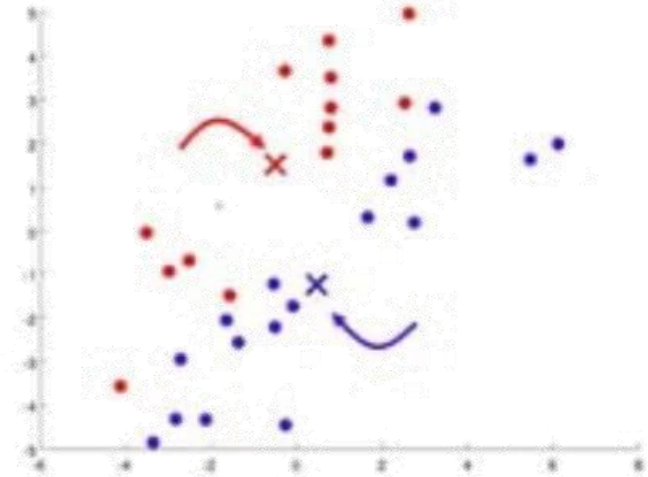
Step 1

All data points are unlabeled.
We randomly initiate two points
called "cluster centroids".



Step 2

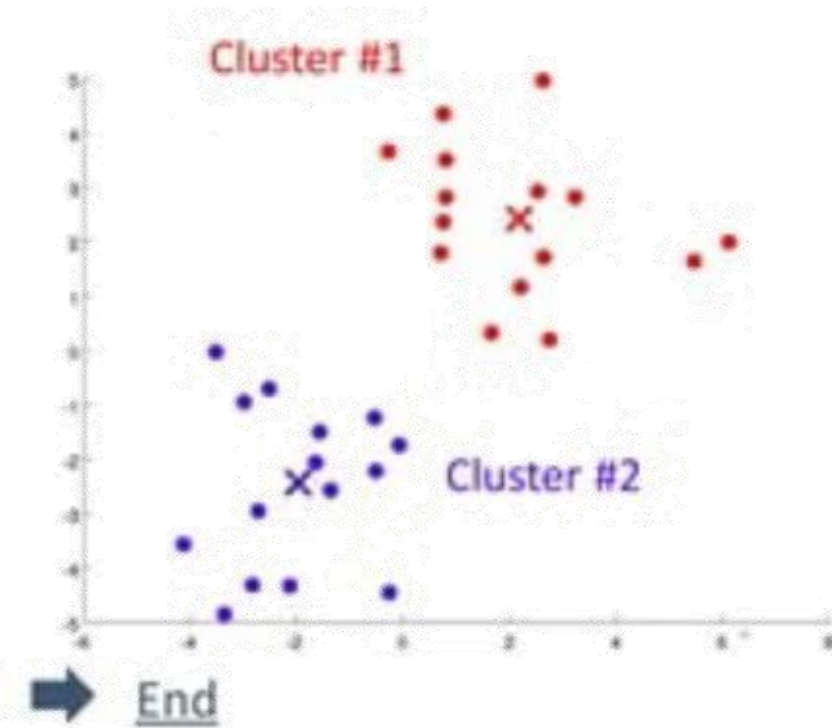
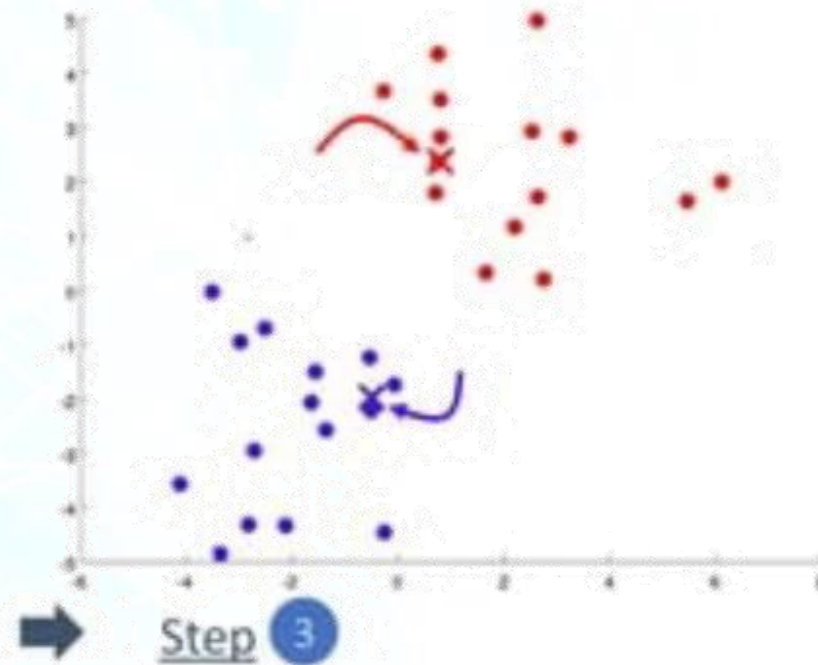
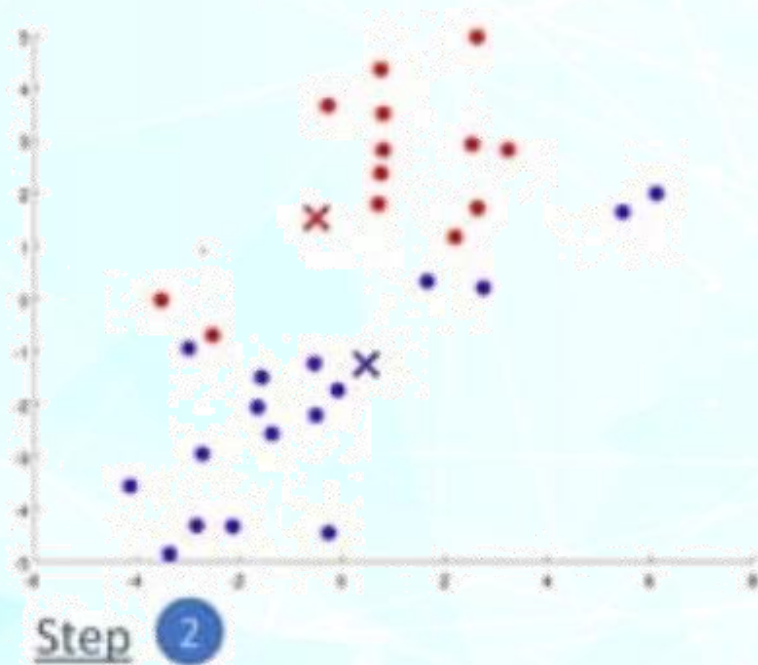
Data points are labeled
according to which centroid
they are the closest from.



Step 3

Each centroid is moved to the
center of the data points that
were labeled in step 2.

Steps 2 and 3 are repeated....



sampai konvergen(tidak ada data yang berpindah label ketika perubahan centroid)



Latihan Code K-NN

X1	X2	y
2	4	0
4	2	0
4	4	0
4	6	1
6	4	1
6	6	1

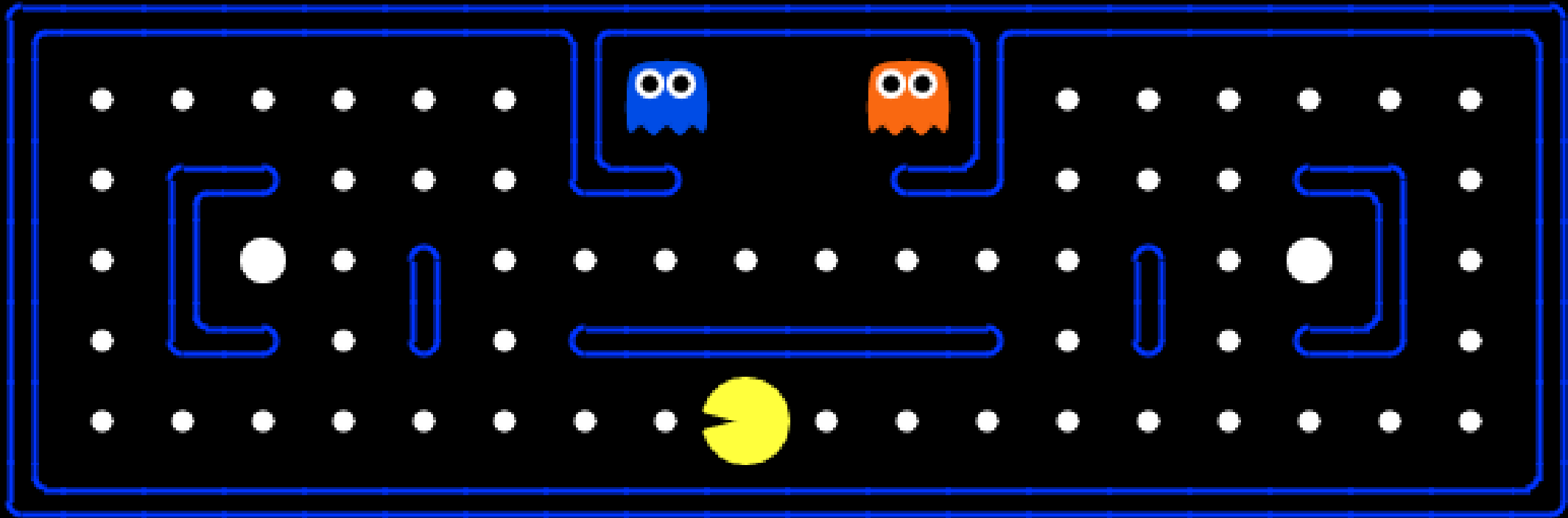
Reinforcement Learning

- Adalah konsep cara belajar mesin untuk mendapatkan suatu keputusan dari suatu interaksi





Reinforcement Learning



SCORE: 0

Eksplorasi dan eksploitasi adalah konsep utama dalam Reinforcement Learning

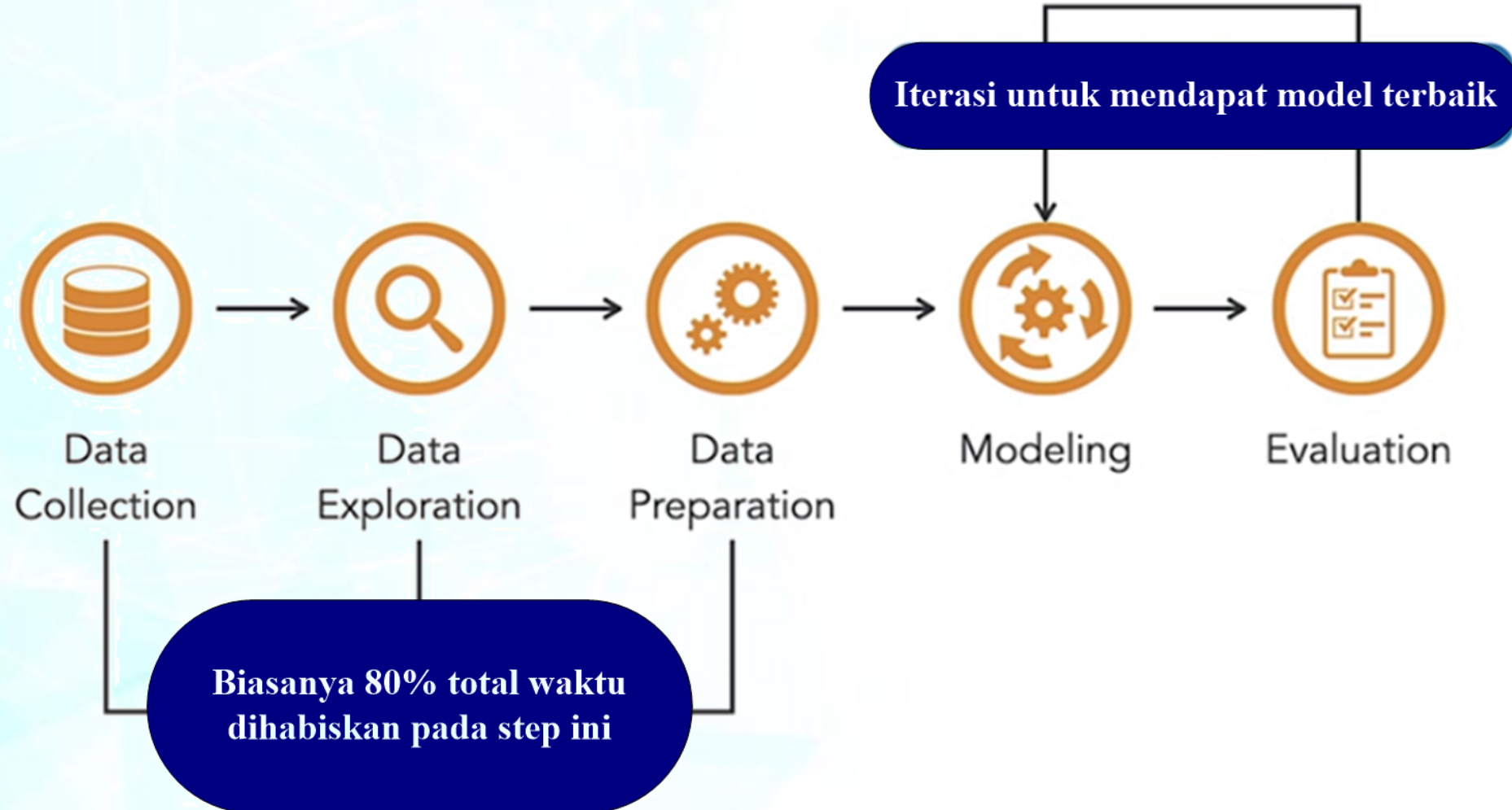
Algoritma Machine Learning



- Pemilihan Algoritma yang tepat
 - Ada puluhan Algoritma supervised and unsupervised machine learning algorithms, masing-masing mengambil pendekatan yang berbeda untuk pembelajarannya
- Tidak ada metode terbaik atau satu ukuran cocok untuk
 - Algoritma yang tepat adalah: just trial and error
 - Bahkan highly experienced data scientists tidak dapat mengatakan apakah sebuah algoritma akan bekerja tanpa dicoba dulu.
- Tetapi pemilihan algoritma juga tergantung pada ukuran dan jenis data yang Anda pakai
 - Wawasan yang Anda ingin mendapatkan dari data
 - Bagaimana wawasan akan digunakan

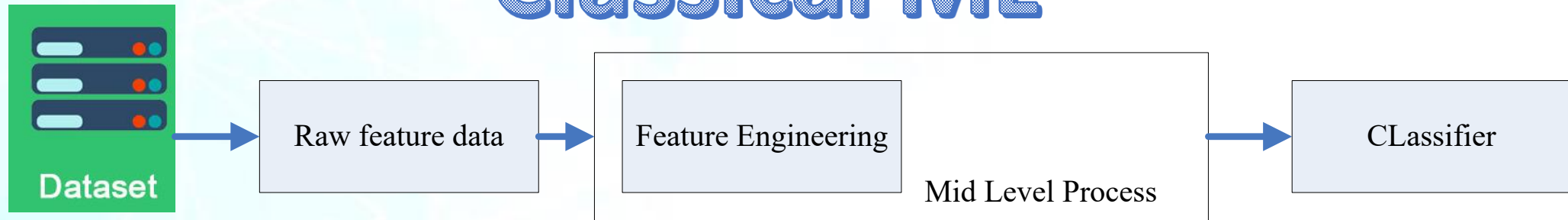


Proses Umum Dalam Machine Learning





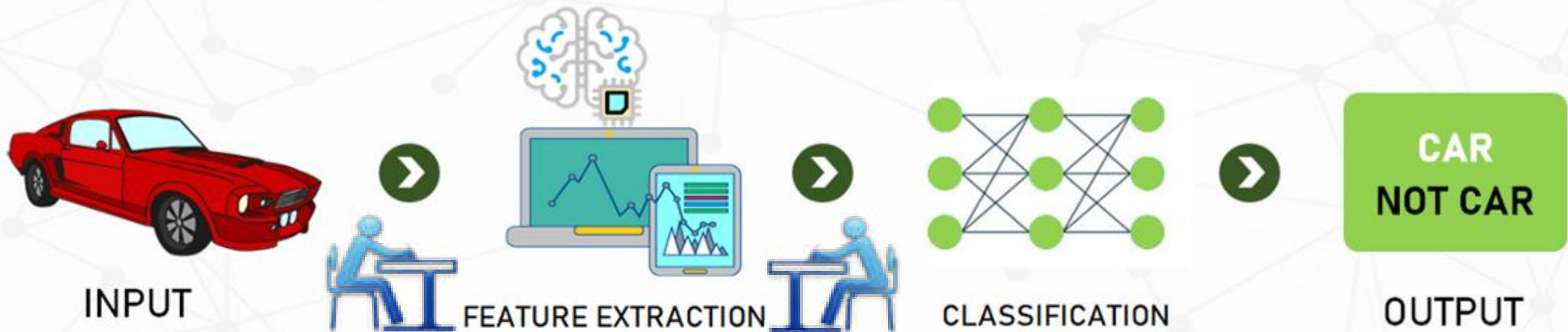
Classical ML



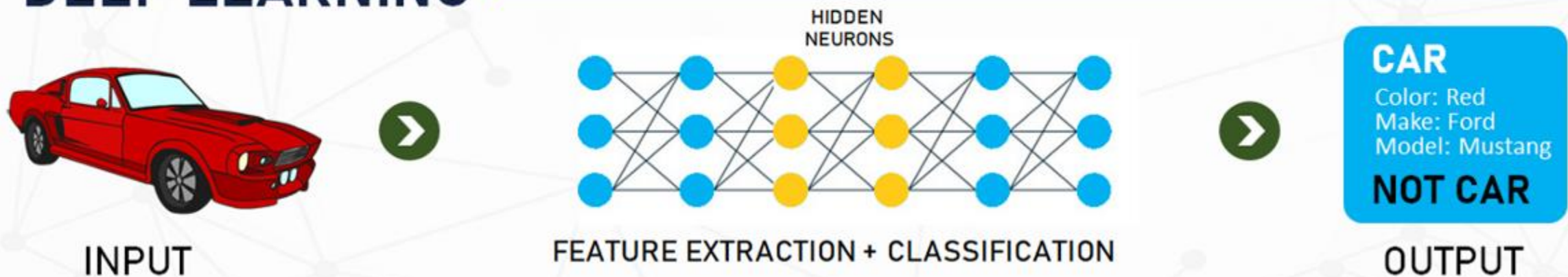
Deep Learning



MACHINE LEARNING



DEEP LEARNING





Resources: Datasets

- UCI Repository: <http://www.ics.uci.edu/~mlearn/MLRepository.html>
- UCI KDD Archive: <http://kdd.ics.uci.edu/summary.data.application.html>
- Statlib: <http://lib.stat.cmu.edu/>
- Delve: <http://www.cs.utoronto.ca/~delve/>



Resources: Journals

- Journal of Machine Learning Research www.jmlr.org
- Machine Learning
- Neural Computation
- Neural Networks
- IEEE Transactions on Neural Networks
- IEEE Transactions on Pattern Analysis and Machine Intelligence
- Annals of Statistics
- Journal of the American Statistical Association
- ...



Resources: Conferences

- International Conference on Machine Learning (ICML)
- European Conference on Machine Learning (ECML)
- Neural Information Processing Systems (NIPS)
- Uncertainty in Artificial Intelligence (UAI)
- Computational Learning Theory (COLT)
- International Joint Conference on Artificial Intelligence (IJCAI)
- International Conference on Neural Networks (Europe)
- ...



**SEE YOU
NEXT WEEK**